LEARNER GUIDE

Platonic Solids

2D and 3D Shapes. A short guide to walk beside you — read as much or as little as helps. *Connection before curriculum. Always.*

What this lesson is about

There are exactly five perfectly regular solids — the Platonic solids: the tetrahedron, cube, octahedron, dodecahedron and icosahedron. In this lesson you meet all five, fold each one up from its flat net, and discover the reason the list stops at five: at every corner the angles must leave a gap. You will also meet Euler's remarkable formula, F + V – E = 2, which holds for every one of them. The question you are exploring is: why are there exactly five perfectly regular solids — no more, no fewer?

What you are learning to notice

- The five Platonic solids, and how to describe each by its faces, vertices and edges.
- That Euler's formula F + V – E = 2 holds for every one of them.
- Why there are exactly five: the angles meeting at a corner must total less than 360° to fold into 3D.
- That the solids come in dual pairs — join the centres of a solid's faces and another Platonic solid appears.

Moving through it, one step at a time

Watch the short film about Euler, then work out F + V – E for each solid in the table. Fold each solid up from its net and rotate it to see the back. In the Corner Investigator, choose a regular polygon and watch whether the angles at a corner leave a gap (a solid), lie flat (a tiling) or overlap (impossible). Then become a solid detective and explore the dual partners. There is no rush; use the isometric grid on the Scratchpad to sketch the solids.

> **A gentle reminder**
> You can pause. You can return. You do not need to complete everything in one sitting.

If you get stuck

Getting stuck is part of doing mathematics, not a sign that something is wrong. There is support built into the lesson, and using it is a strong move:

- **Reconnection Routes** — if an earlier idea feels shaky (the faces, edges and vertices of a solid, the interior angle of a regular polygon, and angles around a point), open a short recap and return whenever you are ready.
- **Socratic prompt** — a question to nudge your thinking.
- **Hint** — a little more help with the step in front of you.
- **Check** — try an answer and see how you are doing; a wrong try is useful information.
- **Worked solution** — a full model to compare your thinking against.

How the worked solution unlocks

The worked solution unlocks after you use the Socratic prompt, use the hint, and check two answers. Those steps are the learning — the worked solution is there to confirm your thinking, not to replace it.

Using support is part of learning. It is not cheating.

Recording your thinking

You can show your working in whatever way suits you today — the on-screen Scratchpad, paper, a spoken explanation, or a photo of a sketch. There is no single right format. If you do not finish everything in one sitting, that is completely fine: what you have done is information for your next step, never a failure.

Before you begin

Settle into a comfortable pace and take the steps one at a time. If your attention drifts, pause and come back — the lesson will keep your place. When you are ready, begin.

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